

Input: 3D Geometric Reconstruction

Object-Level Semantic Perception

Extracts independent semantic object units from raw geometry

- 1 Dense Semantic Segmentation
- 2 3D Object Detection
- 3 Instance & Panoptic Segmentation



Structural & Relational Semantic Modeling

Organizes isolated objects into topological scene structure

- 1 Scene Graph Generation
- 2 Room Layout Estimation



Open-Vocabulary Semantic Understanding

Connects 3D scene entities with language and open-vocabulary concepts

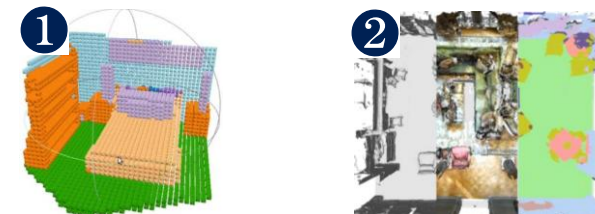
- 1 3D Visual Grounding
- 2 Open-Vocabulary Understanding



Semantic-Geometry Unified Representation

Anchors all semantics onto precise 3D spatial coordinates

- 1 Semantic Occupancy Reconstruction
- 2 Joint Geometry-Semantics Fusion



Output: Structured Semantic Scene Representation